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Field effect transistor with air spacer and method

Abstract

A device includes a substrate, a gate structure, a capping layer, a source/drain region, a source/drain contact, and an air spacer. The gate structure wraps around at least one vertical stack of nanostructure channels over the substrate. The capping layer is on the gate structure. The source/drain region abuts the gate structure. The source/drain contact is on the source/drain region. The air spacer is between the capping layer and the source/drain contact.

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Background/Summary

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits

by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. **1**A-**1**H are diagrammatic top and cross-sectional side views of a portion of an IC device fabricated according to embodiments of the present disclosure.

(3) FIGS. **2**A-**14** are views of various embodiments of an IC device of at various stages of fabrication according to various aspects of the present disclosure.

(4) FIG. **15** is a flowchart illustrating a method of fabricating a semiconductor device according to various aspects of the present disclosure.

DETAILED DESCRIPTION

(5) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(6) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(7) Terms indicative of relative degree, such as “about,” “substantially,” and the like, should be interpreted as one having ordinary skill in the art would in view of current technological norms. Generally, the term “substantially” indicates a tighter tolerance than the term “about.” For example, a thickness of “about 100 units” will include a larger range of values, e.g., 70 units to 130 units (+/−30%), than a thickness of “substantially 100 units,” which will include a smaller range of values, e.g., 95 units to 105 units (+/−5%). Again, such tolerances (+/−30%, +/−5%, and the like) may be process- and/or equipment-dependent, and should not be interpreted as more or less limiting than a person having ordinary skill in the art would recognize as normal for the technology under discussion, other than that “about” as a relative term is not as stringent as “substantially” when used in a similar context.

(8) The present disclosure is generally related to semiconductor devices, and more particularly to field-effect transistors (FETs), such as planar FETs, three-dimensional fin-line FETs (FinFETs), or gate-all-around (GAA) devices. In advanced technology nodes, electronic device performance (e.g., speed) may be sensitive to parasitic capacitance, such as middle-end-of-line (MEOL)

capacitance C.sub.MEOL. For C.sub.MEOL reduction, lower-k dielectric material may reduce capacitance. However, current materials are insufficient in their ability to reduce C.sub.MEOL.
(9) Embodiments of the disclosure provide a process which forms an air spacer by use of a removable metal gate hard mask liner. Embodiments also provide an option to replace the metal gate hard mask (or "self-aligned cap," "SAC") after source/drain contact metal filling. Embodiments are compatible with FinFET, GAAFET, and planar FET devices that include a metal gate hard mask (SAC) process. Air spacer thickness can be controlled by liner deposition.
(10) The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.
(11) FIGS. **1**A-**1**G illustrate diagrammatic cross-sectional side views of a portion of a gate-all-around (GAA) device **20** in accordance with various embodiments. FIGS. **1**A and **1**B illustrate views in an X-Z plane, in which an etch stop layer **131** is partially removed and replaced with an air spacer **131**A. FIGS. **1**C and **1**D illustrate views in the X-Z plane, in which a capping layer **295** (or "SAC") is removed and replaced with a narrower capping layer **1320** and an air spacer **295**A. FIG. **1**E illustrates a view in a Y-Z plane cut through channels **22**A-**22**C, and FIG. **1**F illustrates a view in the Y-Z plane cut through source/drain regions **82**. FIG. **1**G illustrates a view in the Y-Z plan showing a gate via **184** in accordance with various embodiments.
(12) Referring to FIG. **1**A, GAA devices **20**A-**20**C (GAA device **20**C is shown in FIGS. **1**E, **1**F) may be or include one or more N-type FETs (NFETs) or P-type FETs (PFETs). The GAA devices **20**A-**20**C are formed over and/or in a substrate **110**, and generally include gate structures **200** straddling and/or wrapping around semiconductor channels **22**A-**22**D, alternately referred to as "nanostructures," located over semiconductor fins **32** protruding from, and separated by, isolation structures **36** (see FIG. **1**E). The gate structure **200** controls electrical current flow through the channels **22**A-**22**D.
(13) The GAA devices **20**A-**20**C are shown including four channels **22**A-**22**D, which are laterally abutted by source/drain features **82**, and covered and surrounded by the gate structure **200**. Generally, the number of channels **22** is two or more, such as three or four (FIGS. **1**A-**1**F) or more. The gate structure **200** controls flow of electrical current through the channels **22**A-**22**D to and from the source/drain features **82** based on voltages applied at the gate structure **200** and at the source/drain features **82**.
(14) In some embodiments, the fin structure **32** includes silicon. In some embodiments, the GAA devices **20**A-**20**C include an NFET, and the source/drain features **82** thereof include silicon phosphorous (SiP), SiAs, SiSb, SiPAs, SiP:As:Sb, combinations thereof, or the like. In some embodiments, the GAA devices **20**A-**20**C include a PFET, and the source/drain features **82** thereof include silicon germanium (SiGe), either undoped or doped to form, for example, SiGe:B, SiGe:B:Ga, SiGe:Sn, SiGe:B:Sn, or another appropriate semiconductor material. Generally, the source/drain features **82** may include any combination of appropriate semiconductor material(s) and appropriate dopant(s).
(15) The channels **22**A-**22**D each include a semiconductive material, for example silicon or a silicon compound, such as silicon germanium, or the like. The channels **22**A-**22**D are nanostructures (e.g., having sizes that are in a range of a few nanometers) and may also each have an elongated shape and extend in the X-direction. In some embodiments, the channels **22**A-**22**D each have a nanowire (NW) shape, a nanosheet (NS) shape, a nanotube (NT) shape, or other

suitable nanoscale shape. The cross-sectional profile of the channels **22**A-**22**D may be rectangular, round, square, circular, elliptical, hexagonal, or combinations thereof.
(16) In some embodiments, the lengths (e.g., measured in the X-direction) of the channels **22**A-**22**D may be different from each other, for example due to tapering during a fin etching process (see FIGS. **3**A, **3**B). In some embodiments, length of the channel **22**A may be less than a length of the channel **22**B, which may be less than a length of the channel **22**C, which may be less than a length of the channel **22**D. The channels **22**A-**22**D each may not have uniform thickness (e.g., along the X-axis direction), for example due to a channel trimming process used to expand spacing (e.g., measured in the Z-axis direction) between the channels **22**A-**22**D to increase gate structure fabrication process window. For example, a middle portion of each of the channels **22**A-**22**D may be thinner than the two ends of each of the channels **22**A-**22**D. Such shape may be collectively referred to as a “dog-bone” shape, and is illustrated in FIG. **1**A.
(17) In some embodiments, the spacing between the channels **22**A-**22**D (e.g., between the channel **22**B and the channel **22**A or the channel **22**C) is in a range between about 8 nanometers (nm) and about 12 nm. In some embodiments, a thickness (e.g., measured in the Z-direction) of each of the channels **22**A-**22**D is in a range between about 5 nm and about 8 nm. In some embodiments, a width (e.g., measured in the Y-direction, shown in FIG. **1**E, orthogonal to the X-Z plane) of each of the channels **22**A-**22**D is at least about 8 nm.
(18) The gate structure **200** is disposed over and between the channels **22**A-**22**D, respectively. In some embodiments, the gate structure **200** is disposed over and between the channels **22**A-**22**D, which are silicon channels for N-type devices or silicon germanium channels for P-type devices. In some embodiments, the gate structure **200** includes an interfacial layer (IL) **210**, one or more gate dielectric layers **600**, one or more work function tuning layers **900** (see FIG. **14**), and a metal fill layer **290**.
(19) The interfacial layer **210**, which may be an oxide of the material of the channels **22**A-**22**D, is formed on exposed areas of the channels **22**A-**22**D and the top surface of the fin **32**. The interfacial layer **210** promotes adhesion of the gate dielectric layers **600** to the channels **22**A-**22**D. In some embodiments, the interfacial layer **210** has thickness of about 5 Angstroms (A) to about 50 Angstroms (A). In some embodiments, the interfacial layer **210** has thickness of about 10 A. The interfacial layer **210** having thickness that is too thin may exhibit voids or insufficient adhesion properties. The interfacial layer **210** being too thick consumes gate fill window, which is related to threshold voltage tuning and resistance as described above. In some embodiments, the interfacial layer **210** is doped with a dipole, such as lanthanum, for threshold voltage tuning.
(20) In some embodiments, the gate dielectric layer **600** includes at least one high-k gate dielectric material, which may refer to dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide (k≈3.9). Exemplary high-k dielectric materials include HfO.sub.2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, ZrO.sub.2, Ta.sub.2O.sub.5, or combinations thereof. In some embodiments, the gate dielectric layer **600** has thickness of about 5 A to about 100 A.
(21) In some embodiments, the gate dielectric layer **600** may include dopants, such as metal ions driven into the high-k gate dielectric from La.sub.2O.sub.3, MgO, Y.sub.2O.sub.3, TiO.sub.2, Al.sub.2O.sub.3, Nb.sub.2O.sub.5, or the like, or boron ions driven in from B.sub.2O.sub.3, at a concentration to achieve threshold voltage tuning. As one example, for N-type transistor devices, lanthanum ions in higher concentration reduce the threshold voltage relative to layers with lower concentration or devoid of lanthanum ions, while the reverse is true for P-type devices. In some embodiments, the gate dielectric layer **600** of certain transistor devices (e.g., 10 transistors) is devoid of the dopant that is present in certain other transistor devices (e.g., N-type core logic transistors or P-type 10 transistors). In N-type 10 transistors, for example, relatively high threshold voltage is desirable, such that it may be preferable for the 10 transistor high-k dielectric layers to be free of lanthanum ions, which would otherwise reduce the threshold voltage.

(22) In some embodiments, the gate structure **200** further includes one or more work function metal layers, represented collectively as work function metal layer **900**. When configured as an NFET, the work function metal layer **900** of the GAA device **20** may include at least an N-type work function metal layer, an in-situ capping layer, and an oxygen blocking layer. In some embodiments, the N-type work function metal layer is or comprises an N-type metal material, such as TiAlC, TiAl, TaAlC, TaAl, or the like. The in-situ capping layer is formed on the N-type work function metal layer, and may comprise TiN, TiSiN, TaN, or another suitable material. The oxygen blocking layer is formed on the in-situ capping layer to prevent oxygen diffusion into the N-type work function metal layer, which would cause an undesirable shift in the threshold voltage. The oxygen blocking layer may be formed of a dielectric material that can stop oxygen from penetrating to the N-type work function metal layer, and may protect the N-type work function metal layer from further oxidation. The oxygen blocking layer may include an oxide of silicon, germanium, SiGe, or another suitable material. In some embodiments, the work function metal layer **900** includes more or fewer layers than those described.
(23) The work function metal layer **900** may further include one or more barrier layers comprising a metal nitride, such as TiN, WN, MoN, TaN, or the like. Each of the one or more barrier layers may have thickness ranging from about 5 A to about 20 A. Inclusion of the one or more barrier layers provides additional threshold voltage tuning flexibility. In general, each additional barrier layer increases the threshold voltage. As such, for an NFET, a higher threshold voltage device (e.g., an IO transistor device) may have at least one or more than two additional barrier layers, whereas a lower threshold voltage device (e.g., a core logic transistor device) may have few or no additional barrier layers. For a PFET, a higher threshold voltage device (e.g., an IO transistor device) may have few or no additional barrier layers, whereas a lower threshold voltage device (e.g., a core logic transistor device) may have at least one or more than two additional barrier layers. In the immediately preceding discussion, threshold voltage is described in terms of magnitude. As an example, an NFET IO transistor and a PFET IO transistor may have similar threshold voltage in terms of magnitude, but opposite polarity, such as +1 Volt for the NFET IO transistor and −1 Volt for the PFET IO transistor. As such, because each additional barrier layer increases threshold voltage in absolute terms (e.g., +0.1 Volts/layer), such an increase confers an increase to NFET transistor threshold voltage (magnitude) and a decrease to PFET transistor threshold voltage (magnitude).
(24) The gate structure **200** also includes metal fill layer **290**. The metal fill layer **290** may include a conductive material such as Co, W, Ru, combinations thereof, or the like. In some embodiments, the metal fill layer **290** is or includes a Co-, W- or Ru-based compound or alloy including one or more elements, such as Zr, Sn, Ag, Cu, Au, Al, Ca, Be, Mg, Rh, Na, Ir, W, Mo, Zn, Ni, K, Co, Cd, Ru, In, Os, Si, Ge, Mn, combinations thereof, or the like. Between the channels **22**A-**22**D, the metal fill layer **290** is circumferentially surrounded (in the cross-sectional view) by the one or more work function metal layers **900**, which are then circumferentially surrounded by the gate dielectric layers **600**, which are circumferentially surrounded by the interfacial layer **210**. The gate structure **200** may also include a glue layer that is formed between the one or more work function layers **900** and the metal fill layer **290** to increase adhesion. The glue layer is not specifically illustrated in FIGS. **1**A, **1**B for simplicity.
(25) The GAA device **20** may further include source/drain contacts **120** (shown in FIGS. **1**A, **1**B; collectively referred to as “source drain contacts **120**”) that are formed over the source/drain features **82**. The source/drain contacts **120** may include a fill layer **120**F that is or includes a conductive material such as tungsten, ruthenium, cobalt, copper, titanium, titanium nitride, tantalum, tantalum nitride, iridium, molybdenum, nickel, aluminum, or combinations thereof. The source/drain contacts **120** may be surrounded by liner (or, “barrier”) layers **120**L, such as SiN or TiN, which help prevent or reduce diffusion of materials from and into the source/drain contacts **120**. In some embodiments, height of the source/drain contacts **120** may be in a range of about 1

nm to about 50 nm.

(26) Silicide layers **118** are formed between the source/drain features **82** and the source/drain contacts **120**, at least to reduce the source/drain contact resistance. In some embodiments, the silicide layer **118** is or includes TiSi, CrSi, TaSi, MoSi, ZrSi, HfSi, ScSi, Ysi, HoSi, TbSI, GdSi, LuSi, DySi, ErSi, Yb Si, or the like. In some embodiments, the silicide layer **118** is or includes NiSi, CoSi, MnSi, Wsi, FeSi, RhSi, PdSi, RuSi, PtSi, IrSi, OsSi, or the like. The silicide layer **118** may have thickness in a range of about 1 nm to about 10 nm. Thickness lower than about 1 nm may lead to an insufficient reduction in contact resistance. Thickness above about 10 nm may cause electrical shorting with the nanostructures **22**. In some embodiments, the silicide layer **118** is present below, and in contact with, etch stop layer **131**, as shown in FIGS. **1**A-**1**D.

(27) As shown in FIG. **1**F, the GAA device **20** further includes an interlayer dielectric (ILD) **130**. The ILD **130** provides electrical isolation between the various components of the GAA device **20** discussed above, for example between neighboring pairs of the source/drain contacts **120**. An etch stop layer **131**, shown in FIGS. **1**A-**1**D, **1**F, may be formed prior to forming the ILD **130**, and may be positioned laterally between the ILD **130** and the gate spacers **41** and vertically between the ILD **130** and the source/drain features **82**. In some embodiments, the etch stop layer **131** is or includes SiN, SiCN, SiC, SiOC, SiOCN, HfO2, ZrO2, ZrAlOx, HfAlOx, HfSiOx, Al2O3, or other suitable material. In some embodiments, thickness of the etch stop layer is in a range of about 1 nm to about 10 nm. In some embodiments, where the ILD **130** is not present (e.g., is removed completely prior to formation of the source/drain contacts **120**), the etch stop layer **131** may be in contact with the source/drain contact **120**. The etch stop layer **131** may be trimmed, for example, in the X-axis direction prior to formation of the source/drain contact **120** to improve fill quality of the source/drain contact **120**.

(28) Overlying the gate dielectric layer **600** and the gate fill layer **290** are an optional first capping layer **204** (see FIG. **1**E) and a second capping layer **1220** (FIGS. **1**A, **1**B, **1**E) or a second capping layer **1320** (FIGS. **1**C, **1**D). The first capping layer **204** protects the gate structure **200**. In some embodiments, the first capping layer **204** is or includes a dielectric material, such as SiO2, SiN, SiCN, SiOCN, a high-k dielectric material (e.g., Al2O3), or the like. In some embodiments, the first capping layer **204** has thickness (e.g., in the Z-axis direction) in a range of about 1 nm to about 10 nm. The first capping layer **204** may prevent current leakage following one or more etching operations, which may be performed to form gate contacts or gate via **184** (see FIG. **1**G), source/drain contacts **120**, isolation structures (e.g., source/drain contact isolation structures), or the like. In some embodiments, the first capping layer **204** is or comprises a dielectric material that is harder than, for example, the second capping layer **1220**, **1320**, such as aluminum oxide, or other suitable dielectric material.

(29) The second capping layer **1220**, **1320** may be a replacement capping layer, which replaces a second capping layer **295** (see FIG. **11**B, **12**A, **13**A). Each of the second capping layers **295**, **1220**, **1320** may also be referred to as a “self-aligned capping” (SAC) layer, and may provide protection to the underlying gate structure **200**. The second capping layer **295** may act as a CMP stop layer when planarizing the source/drain contacts **120** following formation thereof. The second capping layers **295**, **1220**, **1320** may be dielectric layers including a dielectric material, such as SiO2, SiN, SiCN, SiC, SiOC, SiOCN, HfO2, ZrO2, ZrAlOx, HfAlOx, HfSiOx, Al2O3, BN, or other suitable dielectric material. In some embodiments, over longer gate structures and channels, the second capping layer **295**, **1220**, **1320** may be split by a support structure. In some embodiments, width (X direction) of the second capping layers **295**, **1220**, **1320** is in a range of about 8 nm to about 40 nm.

(30) The GAA devices **20**A-**20**C include gate spacers **41** that are disposed on sidewalls of the gate dielectric layer **600** and the IL **210** above the channel **22**A, and inner spacers **74** that are disposed on sidewalls of the IL **210** between the channels **22**A-**22**D. The inner spacers **74** are also disposed between the channels **22**A-**22**D. The gate spacers **41** and the inner spacers **74** may include a dielectric material, for example a low-k material such as SiOCN, SiON, SiN, or SiOC. In some

embodiments, one or more additional spacer layers are present abutting the gate spacers **41**. The gate spacers **41** may have different height depending on configuration of the air spacers **131**A, **295**A shown in FIGS. **1**A, **1**B or FIGS. **1**C, **1**D, respectively. As shown in FIGS. **1**A, **1**B, the gate spacers **41** extend from upper surfaces of the channels **22**A to lower surfaces of the second capping layers **1220**. Upper surfaces of the gate spacers **41** are generally coplanar with upper surfaces of the gate structures **200** in the configuration shown in FIGS. **1**A, **1**B. In FIGS. **1**C, **1**D, the gate spacers **41** extend from the upper surfaces of the channels **22**A to upper surfaces of the second capping layers **1320** and the source/drain contacts **120**. The upper surfaces of the gate spacers **41** extend past the upper surfaces of the gate structures **200** in the configuration shown in FIGS. **1**C, **1**D.
(31) The air spacers **131**A, **295**A (or "air gaps," or "cavities") are present between the source/drain contacts **120** and the second capping layers **1220**, **1320**, respectively. The air spacers **131**A, **295**A reduce MEOL capacitance C.sub.MEOL, at least by reducing dielectric constant by replacing portions of the ESL **131** or the second capping layer **295**A with air. The reduction in C.sub.MEOL may be as great as 3% or more compared to configurations in which the air spacers **131**A, **295**A are not present. In some embodiments, shown in FIGS. **1**A, **1**C, the air spacers **131**A, **295**A have height in the Z-axis direction that is substantially the same as height of the second capping layers **1220**, **1320**, respectively. The height of the air spacers **131**A, **295**A may be in a range of about 5 nm to about 40 nm. Width of the air spacers **131**A, **295**A may be in a range of about 1 nm to about 10 nm.
(32) In some embodiments, illustrated in FIGS. **1**B, **1**D, seal (or "extension") portions **1231**S, **1331**S of second ESLs **1231**, **1331** overlying the source/drain contacts **120** and the second capping layers **1220**, **1320** extend partially into gaps between the second capping layers **1220**, **1320** and the source/drain contacts **120** (FIG. **1**B) or the spacer layer **41** (FIG. **1**D). In some embodiments, the seal portions **1231**S, **1331**S have height in a range of about 0 nm (FIGS. **1**A, **1**D) to about 10 nm. Width of the seal portions **1231**S, **1331**S may be in a range of about 1 nm to about 10 nm. In some embodiments, the second ESLs **1231**, **1331** including the seal portions **1231**S, **1331**S are or include SiN, SiCN, SiOCN. Second ILDs **1230**, **1330** overlie the second ESLs **1231**, **1331**, respectively, and may be or comprise similar materials as the ILD **130**.
(33) Pullback of gate spacers **41** and ESL **131** may be measured along the X-axis direction (e.g., width) and the Z-axis direction (e.g., height). In FIGS. **1**A, **1**B, the gate spacers **41** and the etch stop layer **131** are pulled back in height by about the same dimension as height of the second capping layer **1220**. In some embodiments, as shown in FIGS. **1**C, **1**D, the gate spacers **41** and the ESL **131** are not pulled back, corresponding to zero reduction in height. Along the X-axis direction, width of the gate spacers **41** and the ESL **131** combined may be pulled back by about 0 nm (FIGS. **1**C, **1**D) to about 15 nm.
(34) FIG. **1**G is a detailed view of the device **10** including a gate via **184** (or "gate contact") in contact with the gate structure **200** of the GAA device **20**A. The gate via **184** is in contact with the metal fill layer **290**, and extends through the second capping layer **1220**, **1320** (the second capping layer **1320** is shown in FIG. **1**G). The gate via **184** extends through the second ILD **1230**, **1330** and the second ESL **1231**, **1331**, as shown. By including the air spacer **295**A or the air spacer **131**A, parasitic capacitance C.sub.MEOL between the gate via **184** and the neighboring source/drain contacts **120** is reduced, which increases speed performance of the GAA device **20**A. The parasitic capacitance C.sub.MEOL may include capacitances C.sub.GMD between the gate via **184** and the source/drain contacts **120**, illustrated in FIG. **1**G. When the gate via **184** is not present over the gate structure **200**, as illustrated in FIG. **1**H, the capacitances C.sub.GMD may be fringe capacitances between the source/drain contacts **120** and the gate structure **200**.
(35) FIG. **1**H illustrates an embodiment in which an air spacer **41**A replaces a portion of the gate spacer **41**. The air spacer **41**A is similar in many respects to the air spacers **295**A, **131**A, which are described with reference to FIGS. **1**A-**1**D. The air spacer **41**A contacts the second capping layer **1220**, the gate spacer **41**, the ESL **131** and the second ESL **1231**. Inclusion of the air spacer **41**A

reduces C.sub.MEOL, such as the fringe capacitances C.sub.GMD, which increases speed of the device **10**.
(36) Additional details pertaining to the fabrication of GAA devices are disclosed in U.S. Pat. No. 10,164,012, titled “Semiconductor Device and Manufacturing Method Thereof” and issued on Dec. 25, 2018, as well as in U.S. Pat. No. 10,361,278, titled “Method of Manufacturing a Semiconductor Device and a Semiconductor Device” and issued on Jul. 23, 2019, the disclosures of each which are hereby incorporated by reference in their respective entireties.
(37) FIG. **15** illustrates a flowchart of a method **1000** for forming an IC device or a portion thereof from a workpiece, according to one or more aspects of the present disclosure. Method **1000** is merely an example and are not intended to limit the present disclosure to what is explicitly illustrated in method **1000**. Additional acts can be provided before, during and after the method **1000** and some acts described can be replaced, eliminated, or moved around for additional embodiments of the method. Not all acts are described herein in detail for reasons of simplicity. Method **1000** is described below in conjunction with fragmentary perspective and/or cross-sectional views of a workpiece, shown in FIGS. **2**A-**14**, at different stages of fabrication according to embodiments of method **1000**. For avoidance of doubt, throughout the figures, the X direction is perpendicular to the Y direction and the Z direction is perpendicular to both the X direction and the Y direction. It is noted that, because the workpiece may be fabricated into a semiconductor device, the workpiece may be referred to as the semiconductor device as the context requires.
(38) FIGS. **2**A through **14** are perspective views and cross-sectional views of intermediate stages in the manufacturing of FETs, such as nanosheet FETs, in accordance with some embodiments. FIGS. **2**A, **3**A, **4**A, **5**A, **6**A, **7**A, **8**A, **9**A and **10**A illustrate perspective views. FIGS. **2**B, **3**B, **4**B, **5**B, **6**B, **7**B, **8**B, **9**B, **10**B and **14** illustrate side views taken along reference cross-section B-B′ (gate cut) shown in FIGS. **2**A, **3**A, **4**A, **5**A and **6**A. FIGS. **4**C, **5**C, **6**C, **6**D, **7**C, **8**C, **9**C, **10**C, **11**A, **11**B, **12**A-**12**I and **13**A-**13**H illustrate side views taken along reference cross-section C-C′ (channel/fin cut) illustrated in FIG. **4**A.
(39) In FIG. **2**A and FIG. **2**B, a substrate **110** is provided. The substrate **110** may be a semiconductor substrate, such as a bulk semiconductor, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The semiconductor material of the substrate **110** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof. Other substrates, such as single-layer, multi-layered, or gradient substrates may be used.
(40) Further in FIG. **2**A and FIG. **2**B, a multi-layer stack **25** or “lattice” is formed over the substrate **110** of alternating layers of first semiconductor layers **21**A-**21**C (collectively referred to as first semiconductor layers **21**) and second semiconductor layers **23**A-**23**C (collectively referred to as second semiconductor layers **23**). In some embodiments, the first semiconductor layers **21** may be formed of a first semiconductor material suitable for n-type nano-FETs, such as silicon, silicon carbide, or the like, and the second semiconductor layers **23** may be formed of a second semiconductor material suitable for p-type nano-FETs, such as silicon germanium or the like. Each of the layers of the multi-layer stack **25** may be epitaxially grown using a process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), vapor phase epitaxy (VPE), molecular beam epitaxy (MBE), or the like.
(41) Three layers of each of the first semiconductor layers **21** and the second semiconductor layers **23** are illustrated. In some embodiments, the multi-layer stack **25** may include one or two each or four or more each of the first semiconductor layers **21** and the second semiconductor layers **23**. Although the multi-layer stack **25** is illustrated as including a second semiconductor layer **23**C as the bottommost layer, in some embodiments, the bottommost layer of the multi-layer stack **25** may

be a first semiconductor layer **21**.

(42) Due to high etch selectivity between the first semiconductor materials and the second semiconductor materials, the second semiconductor layers **23** of the second semiconductor material may be removed without significantly removing the first semiconductor layers **21** of the first semiconductor material, thereby allowing the first semiconductor layers **21** to be patterned to form channel regions of nano-FETs. In some embodiments, the first semiconductor layers **21** are removed and the second semiconductor layers **23** are patterned to form channel regions. The high etch selectivity allows the first semiconductor layers **21** of the first semiconductor material to be removed without significantly removing the second semiconductor layers **23** of the second semiconductor material, thereby allowing the second semiconductor layers **23** to be patterned to form channel regions of nano-FETs.

(43) In FIG. **3**A and FIG. **3**B, fins **32** are formed in the substrate **110** and nanostructures **22**, **24** are formed in the multi-layer stack **25** corresponding to act **1100** of FIG. **15**. In some embodiments, the nanostructures **22**, **24** and the fins **32** may be formed by etching trenches in the multi-layer stack **25** and the substrate **110**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. First nanostructures **22**A-**22**D (also referred to as "channels" below) are formed from the first semiconductor layers **21**, and second nanostructures **24** are formed from the second semiconductor layers **23**. Distance CD**1** between adjacent fins **32** and nanostructures **22**, **24** may be from about 18 nm to about 100 nm. A portion of the device **10** is illustrated in FIGS. **3**A and **3**B including two fins **32** for simplicity of illustration. The process **1000** illustrated in FIGS. **2**A-**14** may be extended to any number of fins, and is not limited to the two fins **32** shown in FIGS. **3**A-**14**.

(44) The fins **32** and the nanostructures **22**, **24** may be patterned by any suitable method. For example, one or more photolithography processes, including double-patterning or multi-patterning processes, may be used to form the fins **32** and the nanostructures **22**, **24**. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing for pitches smaller than what is otherwise obtainable using a single, direct photolithography process. As an example of one multi-patterning process, a sacrificial layer may be formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins **32**.

(45) FIGS. **3**A and **3**B illustrate the fins **32** having tapered sidewalls, such that a width of each of the fins **32** and/or the nanostructures **22**, **24** continuously increases in a direction towards the substrate **110**. In such embodiments, each of the nanostructures **22**, **24** may have a different width and be trapezoidal in shape. In other embodiments, the sidewalls are substantially vertical (non-tapered), such that width of the fins **32** and the nanostructures **22**, **24** is substantially similar, and each of the nanostructures **22**, **24** is rectangular in shape.

(46) In FIGS. **3**A and **3**B, isolation regions **36**, which may be shallow trench isolation (STI) regions, are formed adjacent the fins **32**. The isolation regions **36** may be formed by depositing an insulation material over the substrate **110**, the fins **32**, and nanostructures **22**, **24**, and between adjacent fins **32** and nanostructures **22**, **24**. The insulation material may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by high-density plasma CVD (HDP-CVD), flowable CVD (FCVD), the like, or a combination thereof. In some embodiments, a liner (not separately illustrated) may first be formed along surfaces of the substrate **110**, the fins **32**, and the nanostructures **22**, **24**. Thereafter, a fill material, such as those discussed above may be formed over the liner.

(47) The insulation material undergoes a removal process, such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like, to remove excess insulation material over the nanostructures **22**, **24**. Top surfaces of the nanostructures **22**, **24** may be exposed

and level with the insulation material after the removal process is complete.
(48) The insulation material is then recessed to form the isolation regions **36**. After recessing, the nanostructures **22**, **24** and upper portions of the fins **32** may protrude from between neighboring isolation regions **36**. The isolation regions **36** may have top surfaces that are flat as illustrated, convex, concave, or a combination thereof. In some embodiments, the isolation regions **36** are recessed by an acceptable etching process, such as an oxide removal using, for example, dilute hydrofluoric acid (dHF), which is selective to the insulation material and leaves the fins **32** and the nanostructures **22**, **24** substantially unaltered.
(49) FIGS. **2**A through **3**B illustrate one embodiment (e.g., etch last) of forming the fins **32** and the nanostructures **22**, **24**. In some embodiments, the fins **32** and/or the nanostructures **22**, **24** are epitaxially grown in trenches in a dielectric layer (e.g., etch first). The epitaxial structures may comprise the alternating semiconductor materials discussed above, such as the first semiconductor materials and the second semiconductor materials.
(50) Further in FIG. **3**A and FIG. **3**B, appropriate wells (not separately illustrated) may be formed in the fins **32**, the nanostructures **22**, **24**, and/or the isolation regions **36**. Using masks, an n-type impurity implant may be performed in p-type regions of the substrate **110**, and a p-type impurity implant may be performed in n-type regions of the substrate **110**. Example n-type impurities may include phosphorus, arsenic, antimony, or the like. Example p-type impurities may include boron, boron fluoride, indium, or the like. An anneal may be performed after the implants to repair implant damage and to activate the p-type and/or n-type impurities. In some embodiments, in situ doping during epitaxial growth of the fins **32** and the nanostructures **22**, **24** may obviate separate implantations, although in situ and implantation doping may be used together.
(51) In FIGS. **4**A-**4**C, dummy gate structures **40** are formed over the fins **32** and/or the nanostructures **22**, **24**, corresponding to act **1200** of FIG. **15**. A dummy gate layer **45** is formed over the fins **32** and/or the nanostructures **22**, **24**. The dummy gate layer **45** may be made of materials that have a high etching selectivity versus the isolation regions **36**. The dummy gate layer **45** may be a conductive, semiconductive, or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer **45** may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques for depositing the selected material. A mask layer **47** is formed over the dummy gate layer **45**, and may include, for example, silicon nitride, silicon oxynitride, or the like. In some embodiments, a gate dielectric layer **43** is formed before the dummy gate layer **45** between the dummy gate layer **45** and the fins **32** and/or the nanostructures **22**, **24**. In some embodiments, the mask layer **47** includes a first mask layer **47**A in contact with the dummy gate layer **45**, and a second mask layer **47**B overlying the first mask layer **47**A. The first mask layer **47**A may be or include the same or different material as that of the second mask layer **47**B.
(52) A spacer layer **41** is formed over sidewalls of the mask layer **47** and the dummy gate layer **45**, corresponding to act **1300** of FIG. **15**. The spacer layer **41** is made of an insulating material, such as silicon nitride, silicon oxide, silicon carbo-nitride, silicon oxynitride, silicon oxy carbo-nitride, or the like, and may have a single-layer structure or a multi-layer structure including a plurality of dielectric layers, in accordance with some embodiments. The spacer layer **41** may be formed by depositing a spacer material layer (not shown) over the mask layer **47** and the dummy gate layer **45**. Portions of the spacer material layer between dummy gate structures **40** are removed using an anisotropic etching process, in accordance with some embodiments. In some embodiments, as shown in detail in FIGS. **4**B, **4**C, the spacer layer **41** includes a first spacer layer **41**A in contact with the nanostructure **22**A, the gate dielectric layer **43**, the dummy gate layer **45** and the first and second mask layers **47**A, **47**B. A second spacer layer **41**B of the spacer layer **41** may be in contact with the first spacer layer **41**A and the nanostructure **22**A. The first spacer layer **41**A may be or include the same or different material as that of the second spacer layer **41**B.

(53) FIGS. **4**A-**4**C illustrate one process for forming the spacer layer **41**. In some embodiments, the spacer layer **41** is formed alternately or additionally after removal of the dummy gate layer **45**. In such embodiments, the dummy gate layer **45** is removed, leaving an opening, and the spacer layer **41** may be formed by conformally coating material of the spacer layer **41** along sidewalls of the opening, such as on the ILD **130** (see FIG. **9**A). The conformally coated material may then be removed from the bottom of the opening corresponding to the top surface of the uppermost channel, e.g., the channel **22**A, prior to forming an active gate, such as the gate structure **200**.
(54) While not specifically illustrated in FIGS. **4**A-**4**C, in some embodiments, the hybrid fins **94** are formed following formation of the isolation regions **36** and prior to formation of the dummy gate structures **40**. The hybrid fins **94** may be formed in a self-aligned process by first depositing the liner layer **93** to cover the stacks of nanostructures **22**, **24** shown in FIG. **4**B, then depositing the fill layer **95** to fill remaining portions of openings between the stacks. Excess materials of the liner layer **93** and the fill layer **95** overlying the nanostructures **22**A are then removed, for example, by a planarization process, such as a CMP. If included, the gate isolation structures **99** are then formed over the hybrid fins **94**.
(55) In FIGS. **5**A-**5**C, an etching process is performed to etch the portions of protruding fins **32** and/or nanostructures **22**, **24** that are not covered by dummy gate structures **40**, resulting in the structure shown. The recessing may be anisotropic, such that the portions of fins **32** directly underlying dummy gate structures **40** and the spacer layer **41** are protected, and are not etched. The top surfaces of the recessed fins **32** may be substantially coplanar with the top surfaces of the isolation regions **36** as shown, in accordance with some embodiments. The top surfaces of the recessed fins **32** may be lower than the top surfaces of the isolation regions **36**, in accordance with some other embodiments. FIG. **5**C shows three vertical stacks of nanostructures **22**, **24** following the etching process for simplicity. In general, the etching process may be used to form any number of vertical stacks of nanostructures **22**, **24** over the fins **32**. In some embodiments, the second mask layer **47**B is exposed following the etching process, for example, due to removal of upper portions of the spacer layers **41**A, **41**B during the etching process.
(56) FIGS. **6**A-**6**D illustrate formation of inner spacers **74**, corresponding to act **1300** of FIG. **15**. A selective etching process is performed to recess end portions of the nanostructures **24** exposed by openings in the spacer layer **41** without substantially attacking the nanostructures **22**. After the selective etching process, recesses **64** are formed in the nanostructures **24** at locations where the removed end portions used to be. The resulting structure is shown in FIGS. **6**A, **6**C.
(57) Referring to FIG. **6**D, next, an inner spacer layer is formed to fill the recesses **64** in the nanostructures **22** formed by the previous selective etching process. The inner spacer layer may be a suitable dielectric material, such as silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), or the like, formed by a suitable deposition method such as PVD, CVD, ALD, or the like. An etching process, such as an anisotropic etching process, is performed to remove portions of the inner spacer layers disposed outside the recesses **64**, for example, on sidewalls of the nanostructures **22** and the fins **32**. The remaining portions of the inner spacer layers (e.g., portions disposed inside the recesses **64** in the nanostructures **24**) form the inner spacers **74**. The resulting structure is shown in FIG. **6**D.
(58) FIGS. **7**A-**8**B illustrate formation of source/drain regions **82**. In the illustrated embodiment, the source/drain regions **82** are epitaxially grown from epitaxial material(s). In some embodiments, the source/drain regions **82** exert stress in the respective channels **22**A-**22**D, thereby improving performance. The source/drain regions **82** are formed such that each dummy gate structure **40** is disposed between respective neighboring pairs of the source/drain regions **82**. In some embodiments, the spacer layer **41** separates the source/drain regions **82** from the dummy gate layer **45** by an appropriate lateral distance to prevent electrical bridging to subsequently formed gates of the resulting device.
(59) As described with reference to FIGS. **1**A-**1**F, the source/drain regions **82** may be or include

SiP, SiAs, SiSb, SiPAs, SiP:As:Sb or the like. In some embodiments, the source/drain regions **82** may be or include SiGe:B, SiGe:B:Ga, SiGe:Sn, SiGe:B:Sn or the like. The source/drain regions **82** may exert a compressive strain in the channel regions. The source/drain regions **82** may have surfaces raised from respective surfaces of the fins and may have facets. Neighboring source/drain regions **82** may merge in some embodiments to form a singular source/drain region **82** adjacent two neighboring fins **32**.

(60) In FIGS. **8**A-**8**C, following formation of the source/drain regions **82**, the ILD **130** is formed covering the source/drain regions **82** and abutting the spacer layer **41**. In some embodiments, the ESL **131** is formed prior to forming the ILD **130**. The ESL **131** may be formed by depositing a conformal thin layer of a dielectric material different from that of the ILD **130**, such as one or more of SiN, SiCN, SiC, SiOC, SiOCN, HfO2, ZrO2, ZrAlOx, HfAlOx, HfSiOx, Al2O3, or other suitable material. Following deposition of the ESL **131**, the ILD **130** may be deposited by a suitable process, such as a blanket deposition process, including PVD, CVD, ALD, or the like. The material of the ILD **130** may include silicon dioxide or a low-k dielectric material (e.g., a material having a dielectric constant (k-value) lower than the k-value (about 3.9) of silicon dioxide). The low-k dielectric material may include silicon oxynitride, phosphosilicate glass (PSG), borosilicate glass (BSG), borophosphosilicate glass (BPSG), undoped silicate glass (USG), fluorinated silicate glass (FSG), silicon oxycarbide (SiOxCy), Spin-On-Glass (SOG) or a combination thereof. The ILD **130** may be deposited by spin-on coating, CVD, Flowable CVD (FCVD), PECVD, PVD, or another deposition process.

(61) FIGS. **9**A-**9**D illustrate formation of active gate structures **200**, corresponding to act **1500** of FIG. **15**. A planarization process, such as a chemical mechanical polishing (CMP) process, is performed on the ILD **130** and the ESL **131**. The hard masks **47**A, **47**B and portions of the gate spacers **41** are also removed in the planarization process. After the planarization process, the dummy gate layers **45** are exposed. The top surfaces of the ILD **130** and the ESL **131** may be coplanar with the top surfaces of the dummy gate layers **45** and the gate spacers **41**.

(62) Next, the dummy gate layer **45** is removed in an etching process, so that recesses **92** are formed. In some embodiments, the dummy gate layer **45** is removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gate layer **45** without etching the spacer layer **41**. The dummy gate dielectric **43**, when present, may be used as an etch stop layer when the dummy gate layer **45** is etched. The dummy gate dielectric **43** may then be removed after the removal of the dummy gate layer **45**.

(63) The nanostructures **24** are removed to release the nanostructures **22**. After the nanostructures **24** are removed, the nanostructures **22** form a plurality of nanosheets that extend horizontally (e.g., parallel to a major upper surface of the substrate **110**). In some embodiments, the nanostructures **24** are removed by a selective etching process using an etchant that is selective to the material of the nanostructures **24**, such that the nanostructures **24** are removed without substantially attacking the nanostructures **22**. In some embodiments, the etching process is an isotropic etching process using an etching gas, and optionally, a carrier gas, where the etching gas comprises F2 and HF, and the carrier gas may be an inert gas such as Ar, He, N2, combinations thereof, or the like.

(64) In some embodiments, the nanostructures **24** are removed and the nanostructures **22** are patterned to form channel regions of both PFETs and NFETs. However, in some embodiments the nanostructures **24** may be removed and the nanostructures **22** may be patterned to form channel regions of NFETs, and nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of PFETs. In some embodiments, the nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of NFETs, and the nanostructures **24** may be removed and the nanostructures **22** may be patterned to form channel regions of PFETs. In some embodiments, the nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of both PFETs and NFETs.

(65) In some embodiments, the nanosheets **22** are reshaped (e.g. thinned) by a further etching process to improve gate fill window. The reshaping may be performed by an isotropic etching process selective to the nanosheets **22**. After reshaping, the nanosheets **22** may exhibit the dog bone shape in which middle portions of the nanosheets **22** are thinner than peripheral portions of the nanosheets **22** along the X direction (see FIG. **10**C, for example).
(66) In FIGS. **10**A-**10**C, replacement gates **200** are then formed. The gate structures **200** may be formed by a series of deposition operations, such as ALD cycles, that deposit the various layers of the gate structure **200** in the openings **92**, described below with reference to FIG. **14**.
(67) FIG. **14** is a detailed view of a portion of the gate structure **200**. The gate structure **200** generally includes the interfacial layer (IL, or “first IL” below) **210**, at least one gate dielectric layer **600**, the work function metal layer **900**, and the gate fill layer **290**. In some embodiments, each replacement gate **200** further includes at least one of a second interfacial layer **240** or a second work function layer **700**.
(68) With reference to FIG. **14**, in some embodiments, the first IL **210** includes an oxide of the semiconductor material of the substrate **110**, e.g. silicon oxide. In other embodiments, the first IL **210** may include another suitable type of dielectric material. The first IL **210** has a thickness in a range between about 5 angstroms and about 50 angstroms.
(69) Still referring to FIG. **14**, the gate dielectric layer **600** is formed over the first IL **210**. In some embodiments, an atomic layer deposition (ALD) process is used to form the gate dielectric layer **600** to control thickness of the deposited gate dielectric layer **600** with precision. In some embodiments, the ALD process is performed using between about 40 and 80 deposition cycles, at a temperature range between about 200 degrees Celsius and about 300 degrees Celsius. In some embodiments, the ALD process uses HfCl4 and/or H2O as precursors. Such an ALD process may form the first gate dielectric layer **220** to have a thickness in a range between about 10 angstroms and about 100 angstroms.
(70) In some embodiments, the gate dielectric layer **600** includes a high-k dielectric material, which may refer to dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide (k≈3.9). Exemplary high-k dielectric materials include HfO.sub.2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, ZrO.sub.2, Ta.sub.2O.sub.5, or combinations thereof. In other embodiments, the gate dielectric layer **600** may include a non-high-k dielectric material such as silicon oxide. In some embodiments, the gate dielectric layer **600** includes more than one high-k dielectric layer, of which at least one includes dopants, such as lanthanum, magnesium, yttrium, or the like, which may be driven in by an annealing process to modify threshold voltage of the GAA devices **20**A-**20**E.
(71) With further reference to FIG. **14**, the second IL **240** is formed on the gate dielectric layer **600**, and the second work function layer **700** is formed on the second IL **240**. The second IL **240** promotes better metal gate adhesion on the gate dielectric layer **600**. In many embodiments, the second IL **240** further provides improved thermal stability for the gate structure **200**, and serves to limit diffusion of metallic impurity from the work function metal layer **900** and/or the work function barrier layer **700** into the gate dielectric layer **600**. In some embodiments, formation of the second IL **240** is accomplished by first depositing a high-k capping layer (not illustrated for simplicity) on the gate dielectric layer **600**. The high-k capping layer comprises one or more of the following: HfSiON, HfTaO, HfTiO, HfTaO, HfAlON, HfZrO, or other suitable materials, in various embodiments. In a specific embodiment, the high-k capping layer comprises titanium silicon nitride (TiSiN). In some embodiments, the high-k capping layer is deposited by an ALD using about 40 to about 100 cycles at a temperature of about 400 degrees C. to about 450 degrees C. A thermal anneal is then performed to form the second IL **240**, which may be or comprise TiSiNO, in some embodiments. Following formation of the second IL **240** by thermal anneal, an atomic layer etch (ALE) with artificial intelligence (AI) control may be performed in cycles to remove the high-k capping layer while substantially not removing the second IL **240**. Each cycle

may include a first pulse of WCl.sub.5, followed by an Ar purge, followed by a second pulse of O.sub.2, followed by another Ar purge. The high-k capping layer is removed to increase gate fill window for further multiple threshold voltage tuning by metal gate patterning.
(72) Further in FIG. **14**, after forming the second IL **240** and removing the high-k capping layer, the work function barrier layer **700** is optionally formed on the gate structure **200**, in accordance with some embodiments. The work function barrier layer **700** is or comprises a metal nitride, such as TiN, WN, MoN, TaN, or the like. In a specific embodiment, the work function barrier layer **700** is TiN. The work function barrier layer **700** may have thickness ranging from about 5 A to about 20 A. Inclusion of the work function barrier layer **700** provides additional threshold voltage tuning flexibility. In general, the work function barrier layer **700** increases the threshold voltage for NFET transistor devices, and decreases the threshold voltage (magnitude) for PFET transistor devices.
(73) The work function metal layer **900**, which may include at least one of an N-type work function metal layer, an in-situ capping layer, or an oxygen blocking layer, is formed on the work function barrier layer **700**, in some embodiments. The N-type work function metal layer is or comprises an N-type metal material, such as TiAlC, TiAl, TaAlC, TaAl, or the like. The N-type work function metal layer may be formed by one or more deposition methods, such as CVD, PVD, ALD, plating, and/or other suitable methods, and has a thickness between about 10 A and 20 A. The in-situ capping layer is formed on the N-type work function metal layer. In some embodiments, the in-situ capping layer is or comprises TiN, TiSiN, TaN, or another suitable material, and has a thickness between about 10 A and 20 A. The oxygen blocking layer is formed on the in-situ capping layer to prevent oxygen diffusion into the N-type work function metal layer, which would cause an undesirable shift in the threshold voltage. The oxygen blocking layer is formed of a dielectric material that can stop oxygen from penetrating to the N-type work function metal layer, and may protect the N-type work function metal layer from further oxidation. The oxygen blocking layer may include an oxide of silicon, germanium, SiGe, or another suitable material. In some embodiments, the oxygen blocking layer is formed using ALD and has a thickness between about 10 A and about 20 A.
(74) FIG. **14** further illustrates the metal fill layer **290**. In some embodiments, a glue layer (not separately illustrated) is formed between the oxygen blocking layer of the work function metal layer and the metal fill layer **290**. The glue layer may promote and/or enhance the adhesion between the metal fill layer **290** and the work function metal layer **900**. In some embodiments, the glue layer may be formed of a metal nitride, such as TiN, TaN, MoN, WN, or another suitable material, using ALD. In some embodiments, thickness of the glue layer is between about 10 A and about 25 A. The metal fill layer **290** may be formed on the glue layer, and may include a conductive material such as tungsten, cobalt, ruthenium, iridium, molybdenum, copper, aluminum, or combinations thereof. In some embodiments, the metal fill layer **290** may be deposited using methods such as CVD, PVD, plating, and/or other suitable processes. In some embodiments, a seam **510**, which may be an air gap, is formed in the metal fill layer **290** vertically between the channels **22**A, **22**B. In some embodiments, the metal fill layer **290** is conformally deposited on the work function metal layer **900**. The seam **510** may form due to sidewall deposited film merging during the conformal deposition. In some embodiments, the seam **510** is not present between the neighboring channels **22**A, **22**B.
(75) In FIGS. **11**A, **11**B, following formation of the gate structures **200**, an optional first capping layer **204** and a second capping layer **295** are formed overlying the gate dielectric layer **600** and the gate fill layer **290**. The first capping layer **204** protects the gate structure **200**. Prior to forming the capping layers **204**, **295**, the gate structures **200** may be recessed by, for example, a suitable etching operation (e.g., ALE), as shown in FIG. **11**A.
(76) As illustrated in FIG. **11**B, in some embodiments, the first capping layer **204** is formed by depositing a dielectric material, such as SiO2, SiN, SiCN, SiOCN, a high-k dielectric material (e.g., Al2O3), or the like, by a suitable deposition process, such as PVD, CVD, ALD, or the like. The

first capping layer **204** may prevent current leakage following one or more etching operations, which may be performed to form gate contacts, source/drain contacts **120**, isolation structures (e.g., source/drain contact isolation structures), or the like. In some embodiments, the first capping layer **204** is or comprises a dielectric material that is harder than, for example, the second capping layer **295**, such as aluminum oxide, or other suitable dielectric material. The first capping layer **204** is not shown in subsequent figures for simplicity of illustration.

(77) The second capping layer **295**, also referred to as a "self-aligned capping" (SAC) layer, may provide protection to the underlying gate structure **200**, and may also act as a CMP stop layer when planarizing the source/drain contacts **120** following formation thereof. The second capping layer **295** may be formed by depositing a dielectric layer including a dielectric material, such as SiO2, SiN, SiCN, SiC, SiOC, SiOCN, HfO2, ZrO2, ZrAlOx, HfAlOx, HfSiOx, Al2O3, BN, or other suitable dielectric material, by a suitable deposition process, such as PVD, CVD, ALD, or the like. In some embodiments, over longer gate structures and channels, the second capping layer **295** may be split by a support structure. In some embodiments, width (X direction) of the second capping layer **295** is in a range of about 8 nm to about 40 nm.

(78) In FIGS. **12**A-**12**I, the air spacer **131**A is formed, corresponding to acts **1700-2300** of FIG. **15**. In FIG. **12**A, following formation of the second capping layer **295**, one or more removal operations, such as an atomic layer etch (ALE), are performed to remove portions of the ILD **130** and the ESL **131** overlying the source/drain regions **82**. Following the removal operations, upper surfaces of the source/drain regions **82** are exposed. In some embodiments, upper regions of the source/drain regions **82** are recessed slightly in the removal operations, such that the upper surfaces are convex, as shown in FIG. **12**A.

(79) In FIG. **12**B, silicide regions **118** and the source/drain contacts **120** are formed on the source/drain regions **82**, corresponding to act **1700** of FIG. **15**. In some embodiments, P-type source/drain regions **82** are implanted with dopants, which may be optionally followed by an anneal. The dopants may include P-type implants, such as Ga, B, C, Sn or the like, implanted to a depth in a range of about 3 nm to about 10 nm, at a concentration in a range of about 10.sup.18 cm.sup.−3 to about 10.sup.21 cm.sup.−3. Implanting the dopants improves activation of the P-type source/drain regions **82**.

(80) In some embodiments, the silicide layers **118** are formed prior to formation of the source/drain contacts **120**. For example, an N-type or P-type metal layer may be formed as a conformal thin layer over exposed portions of the source/drain regions **82**. The metal layer may be or include one or more of Ni, Co, Mn, W, Fe, Rh, Pd, Ru, Pt, Ir, Os or the like. In some embodiments, the metal layer is or includes one or more of Ti, Cr, Ta, Mo, Zr, Hf, Sc, Ys, Ho, Tb, Gd, Lu, Dy, Er, Yb or another suitable material. Following formation of the metal layer, the silicide layers **118** may be formed by annealing the device **10** with the metal layer **115** in contact with the source/drain regions **82**. Following the anneal, the silicide layers **118** may be or include one or more of NiSi, CoSi, MnSi, Wsi, FeSi, RhSi, PdSi, RuSi, PtSi, IrSi, OsSi, TiSi, CrSi, TaSi, MoSi, ZrSi, HfSi, ScSi, Ysi, HoSi, TbSI, GdSi, LuSi, DySi, ErSi, YbSi or the like. Silicide of the silicide layers **118** may diffuse into regions below the ESL **131**, as shown in FIG. **12**B. Thickness of the silicide layers **118** may be in a range of about 1 nm to about 10 nm. Below about 1 nm, contact resistance may be too high. Above about 10 nm, the silicide layers **118** may short with the channels **22**A. In some embodiments, the P+ implanted region of the P-type source/drain region **82** may not react completely with the metal layer when forming the silicide layers **118**. As such, a P+ doped region may remain under the silicide layers **118**.

(81) In FIG. **12**B, following formation of the silicide layers **118**, unreacted portions of the metal layer are removed. Following removal of the metal layer, the source/drain contacts **120** are formed by filling the openings over the source/drain regions **82** with, for example, the liner layer **120**L and the fill layer **120**L. In some embodiments, the source/drain contacts **120** are formed by depositing a material that is or includes a conductive material such as Co, W, Ru, combinations thereof, or the

like. In some embodiments, the source/drain contacts **120** are or include a Co-, W- or Ru-based compound or alloy including one or more elements, such as Zr, Sn, Ag, Cu, Au, Al, Ca, Be, Mg, Rh, Na, Ir, W, Mo, Zn, Ni, K, Co, Cd, Ru, In, Os, Si, Ge, Mn, combinations thereof, or the like. The source/drain contacts **120** land on the silicide layer **118** and are in contact with the ESL **131** and the hybrid fins **94**. Description of the device **10** and illustration thereof in many of the figures is given with reference to GAAFETs including vertical stacks of the nanostructures **22**. In some embodiments, the silicide layers **118** and the source/drain contacts **120** are formed in and on source/drain regions **82** of FinFET devices.

(82) Further to FIG. **12**B, following formation of the source/drain contacts **120** and the silicide layers **118**, the second capping layers **295** are removed (corresponding to act **1800** of FIG. **15**) in preparation for forming the air spacers **131**A in subsequent processes. The second capping layers **295** may be removed by a suitable etching process, such as ALE, using an etchant that removes the second capping layers **295** without substantially attacking the spacer layer **41** or the gate structure **200**.

(83) In FIG. **12**C, following formation of the silicide layers **118** and the source/drain contacts **120**, and removal of the second capping layers **295**, the spacer layer **41** and the ESL **131** are recessed to a level substantially coplanar with upper surfaces of the gate structures **200**, as shown, corresponding to act **1900** of FIG. **15**. In some embodiments, the spacer layer **41** is recessed (or “pulled back”) in a first etching operation, such as an ALE, using an etchant that removes the spacer layer **41** without substantially attacking the ESL **131**. Following recessing of the spacer layer **41**, the ESL **131** is recessed in a second etching operation, such as an ALE, such that heights of the spacer layer **41** and the ESL **131** are substantially the same, and upper surfaces of the gate structure **200**, the spacer layer **41** and the ESL **131** are substantially coplanar. In some embodiments, one or more of the spacer layer **41** and the ESL **131** may have an upper surface that is at a level higher than the upper surface of the gate structure **200** following the first and second etching operations. Act **1900** is optional, and embodiments in which act **1900** is omitted are described with reference to FIGS. **13**A-**13**H. In some embodiments, the spacer layer **41** is pulled back, and the ESL **131** is not substantially pulled back, which leads to formation of the structure shown in FIG. **1**H.

(84) In FIG. **12**D, following recessing of the gate structure **200**, the spacer layer **41** and the ESL **131**, a dummy liner layer **1210**, which may be a removable hard mask liner layer, is formed as a conformal thin layer over exposed surfaces of the gate structure **200**, the spacer layer **41**, the ESL **131** and the source/drain contact **120**, corresponding to act **2000** of FIG. **15**. In some embodiments, the dummy liner layer **1210** is or includes one or more of SiN, Si, SiGe, SiOx or the like. In some embodiments, the dummy liner layer **1210** includes a different material than the gate dielectric layer **600** and the spacer layer **41**. The dummy liner layer **1210** may be deposited by a suitable deposition process, such as PVD, CVD, ALD or the like. Thickness of the dummy liner layer **1210** may be in a range of about 1 nm to about 10 nm, and may correspond to thickness of the ESL **131**. For example, the thickness of the dummy liner layer **1210** may be substantially the same as the thickness of the ESL **131**. In some embodiments, the dummy liner layer **1210** is slightly thicker or slightly thinner than the ESL **131**. In embodiments in which the ESL **131** is not substantially pulled back, the dummy liner layer **1210** may be formed on sidewalls of the ESL **131**.

(85) In FIG. **12**E, following formation of the dummy liner layer **1210**, horizontal portions of the dummy liner layer **1210** overlying the spacer layer **41** and the gate electrode **200** are removed. The removal may be by an appropriate etching process, such as an anisotropic etching process, which may be or include an ALE. Following removal of the horizontal portions of the dummy liner layer **1210**, upper surfaces of the gate structure **200** and the spacer layer **41** are exposed.

(86) In FIG. **12**F, following exposing the upper surfaces of the gate structure **200** and the spacer layer **41** by removing the horizontal portions of the dummy liner layer **1210**, the second capping layer **1220** is formed on exposed surfaces of, and filling the opening between and over, the dummy

liner layer **1210**, the gate structure **200** and the spacer layer **41**, corresponding to act **2100** of FIG. **15**. The second capping layer **1220** may be formed by an appropriate deposition process, such as a PVD, CVD or ALD, or the like. In some embodiments, the second capping layer **1220** is or includes SiN, SiCN, SiCON, SiOx, an extreme low-k dielectric (k<2), another suitable low-k dielectric (e.g., 2<k<3.9) or the like. In some embodiments, the second capping layer **1220** is an oxide of silicon, such as SiO.sub.2. The second capping layer **1220** generally includes a different material than the dummy liner layer **1210**, such as a material having a different etch selectivity than that of the dummy liner layer **1210**. The second capping layer **1220** may be formed of a material having lower dielectric constant than the second capping layer **295**, which lowers C.sub.MEOL. For example, the second capping layer **295** may be a harder material, which has higher dielectric constant, to provide better protection to the gate structure **200** during formation of the source/drain contacts **120**. The second capping layer **1220**, which replaces the second capping layer **295**, is not generally used for protection during formation of nearby metal layers, as the source/drain contacts **120** are already formed. As such, the second capping layer **1220** has greater flexibility in choice of material, and a lower dielectric constant material may be chosen that reduces C.sub.MEOL.

(87) In FIG. **12**G, following formation of the second capping layer **1220**, the dummy liner layer **1210** is removed, corresponding to act **2200** of FIG. **15**. Removal of the dummy liner layer **1210** forms the air spacer **131**A is in the second capping layer **1220** and the source/drain contact **120**. In some embodiments, the dummy liner layer **1210** is removed by an etching process, such as an ALE. Removal of the dummy liner layer **1210** may stop at an upper surface of the ESL **131**. Formation of the air spacer **131**A, which may also be considered replacing a portion of the ESL **131** with the air spacer **131**A, reduces middle-end-of-line capacitance, as the dielectric constant of the air spacer **131**A is lower than that of the ESL **131**. In embodiments in which the ESL **131** is not pulled back, the air spacer **41**A (see FIG. **1**H) may be formed by removing the dummy liner layer **1210**.

(88) In FIGS. **12**H, **12**I, the second ESL **1231** and the second ILD **1230** are formed following formation of the air spacer **131**A, corresponding to act **2300** of FIG. **15**. The second ESL **1231** and the second ILD **1230** may be formed by appropriate deposition processes such as PVD, CVD, ALD or the like. The second ESL **1230** may be formed by depositing a conformal thin layer of a dielectric material different from that of the second ILD **1230**, such as one or more of SiN, SiCN, SiC, SiOC, SiOCN, HfO2, ZrO2, ZrAlOx, HfAlOx, HfSiOx, Al2O3, or other suitable material. Following deposition of the second ESL **1231**, the second ILD **1230** may be deposited by a suitable process, such as a blanket deposition process, including PVD, CVD, ALD, or the like. The material of the second ILD **1230** may include silicon dioxide or a low-k dielectric material (e.g., a material having a dielectric constant (k-value) lower than the k-value (about 3.9) of silicon dioxide). The low-k dielectric material may include silicon oxynitride, phosphosilicate glass (PSG), borosilicate glass (BSG), borophosphosilicate glass (BPSG), undoped silicate glass (USG), fluorinated silicate glass (FSG), silicon oxycarbide (SiOxCy), Spin-On-Glass (SOG) or a combination thereof. The second ILD **1230** may be deposited by spin-on coating, CVD, Flowable CVD (FCVD), PECVD, PVD, or another deposition process. FIG. **12**H illustrates an embodiment in which the second ESL **1231** does not significantly extend into the lateral space between the second capping layer **1220** and the source/drain contact **120**. FIG. **12**I illustrates an embodiment in which the second ESL **1231** extends partially into the lateral space between the second capping layer **1220** and the source/drain contact **120**. Extension portions **1231**S, which may also be referred to as “seal portions **1231**S,” of the second ESL **1231** may have width substantially the same as that of the air spacer **131**A, and height in a range of about 0 nm (corresponding to FIG. **12**H) to about 10 nm. The extension portions **1231**S are in contact with sidewalls of the source/drain contacts **120**, such as the liner layers **120**L, and sidewalls of the second capping layers **1220**. Lower surfaces of the extension portions **1231**S are separated from the gate structures **200** by the air spacers **295**A.

(89) FIGS. **13**A-**13**H illustrate formation of the air spacer **295**A according to another embodiment. Many of the operations illustrated in FIGS. **13**A-**13**H are similar to those described with reference

to FIGS. **12**A-**12**I.

(90) FIGS. **13**A and **13**B are substantially the same as FIGS. **12**A and **12**B. Formation of the source/drain contacts **120**, recessing of the gate structure **200** and removal of the second capping layer **295** illustrated in FIGS. **13**A, **13**B are described above with reference to FIGS. **12**A, **12**B, and not repeated here.

(91) In FIG. **13**C, following recessing of the gate structure **200** and removal of the second capping layer **295**, corresponding to act **1700**, **1800** of FIG. **15**, the optional act **1900** is omitted, and the dummy liner layer **1310**, which is substantially the same as the dummy liner layer **1210**, is formed on exposed surfaces of the spacer layer **41** and the gate structure **200**. FIG. **13**C is similar in many respects to FIG. **12**D. In FIG. **13**C, because the spacer layer **41** and the ESL **131** are not recessed, the dummy liner layer **1310** is formed on vertical sidewalls of the spacer layer **41**, whereas the dummy liner layer **1210** is formed on upper surfaces of the spacer layer **41**. Formation of the dummy liner layer **1310** is substantially the same as formation of the dummy liner layer **1210**, and description thereof is provided with reference to FIG. **12**D. FIG. **13**D is similar in many respects to FIG. **12**E, and illustrates removal of horizontal portions of the dummy liner layer **1310** overlying the gate structure **200**, which is substantially described with reference to FIG. **12**E, and not repeated here.

(92) Formation of the second capping layer **1320** is illustrated in FIG. **13**E, and formation of the air spacer **295**A is illustrated in FIG. **13**F. FIG. **13**E is similar in many respects to FIG. **12**F, and FIG. **13**F is similar in many respects to FIG. **12**G. The air spacer **295**A may have dimensions that are similar to those of the air spacer **131**A. The air spacer **295**A is present between the spacer layer **41** the second capping layer **1320** and the gate structure **200**. Dielectric constant of the air spacer **295**A is less than that of the second capping layer **1320**, such that the air spacer **295**A reduces middle-end-of-line capacitance compared to configurations in which the space between the spacer layers **41** is completely filled by the second capping layer **295** or the second capping layer **1320**.

(93) FIGS. **13**G, **13**H are similar in many respects to FIGS. **12**H, **12**I, respectively, and illustrate formation of the second ESL **1331** and the second ILD **1330** in accordance with various embodiments. In the embodiment shown in FIG. **13**G, the second ESL **1331** does not significantly extend into the lateral space between the spacer layer **41** and the second capping layer **1320** occupied by the air spacer **295**A. In the embodiment shown in FIG. **13**H, the second ESL **1331** extends partially into the space between the spacer layer **41** and the second capping layer **1320**. Extension portions **1331**S of the second ESL **1331**, which may also be referred to as “seal portions **1331**S,” are similar in many respects to the extension portions **1231**S described with reference to FIG. **12**I. The extension portions **1331**S are in contact with sidewalls of the spacer layers **41** and sidewalls of the second capping layers **1320**. Lower surfaces of the extension portions **1331**S are separated from the gate structures **200** by the air spacers **295**A. Sidewalls of the extension portions **1331**S are separated from sidewalls of the liner layer **120**L by the spacer layer **41** and the ESL **131**.

(94) Additional processing may be performed to finish fabrication of the GAA devices **20**. For example, gate contacts (or gate vias, e.g., the gate via **184**) may be formed to electrically couple to the gate structures **200**. An interconnect structure may then be formed over the source/drain contacts **120** and the gate contacts. The interconnect structure may include a plurality of dielectric layers (including, for example, the second ILD **1230**, **1330**) surrounding metallic features, including conductive traces and conductive vias, which form electrical connection between devices on the substrate **110**, such as the GAA devices **20**, as well as to IC devices external to the IC device **10**. In some embodiments, additional capping layers (not shown) are present over the source/drain contacts **120**. Configurations in which only the second capping layers **1220**, **1320** over the gate structures **200** are present (e.g., no second capping layers are present over the source/drain contacts **120**) may be considered “single SAC” structures, and configurations in which the second capping layers **1220**, **1320** and the additional capping layers over the gate structures **200** are both present may be considered “double SAC” structures.

(95) Embodiments may provide advantages. Speed performance of devices including the air spacers **131**A, **295**A is increased by middle-end-of-line capacitance C.sub.MEOL reductions due to the low dielectric constant of the air spacers **131**A, **295**A which replace higher dielectric constant materials of the ESL **131** or the second capping layer **295**, respectively. Replacement of the second capping layer **295** with a lower-k dielectric material of the second capping layers **1220**, **1320** also reduces C.sub.MEOL.
(96) In accordance with at least one embodiment, a device includes a substrate, a gate structure, a capping layer, a source/drain region, a source/drain contact, and an air spacer. The gate structure wraps around at least one vertical stack of nanostructure channels over the substrate. The capping layer is on the gate structure. The source/drain region abuts the gate structure. The source/drain contact is on the source/drain region. The air spacer is between the capping layer and the source/drain contact.
(97) In accordance with at least one embodiment, a device includes: a substrate; a first source/drain contact over the substrate; a second source/drain contact laterally offset from the first source/drain contact; a capping layer laterally between the first and second source/drain contacts; a first air spacer between the first source/drain contact and the capping layer; and a second air spacer between the second source/drain contact and the capping layer.
(98) In accordance with at least one embodiment, a method includes: forming a gate structure over and wrapping around a vertical stack of nanostructures over a substrate; forming a first capping layer on the gate structure; removing the first capping layer; forming a liner layer on vertical sidewalls of source/drain contacts on either side of the gate structure; forming a second capping layer on the gate structure between the liner layer; and forming air spacers by removing the liner layer.
(99) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device, comprising: a substrate; a gate structure wrapping around at least one vertical stack of nanostructure channels over the substrate; a capping layer on the gate structure; a source/drain region abutting the gate structure; a source/drain contact on the source/drain region; and an air spacer positioned laterally between the capping layer and the source/drain contact, wherein the air spacer is entirely located within a vertical region defined by a lower boundary that is at or above a top surface of the gate structure and an upper boundary that is at or below a top surface of the capping layer.
2. The device of claim 1, comprising: a spacer layer between the air spacer and the capping layer.
3. The device of claim 1, comprising: a spacer layer below the air spacer and beside the gate structure.
4. The device of claim 1, comprising: a dielectric layer over the capping layer and the air spacer.
5. The device of claim 4, wherein the dielectric layer includes: an extension portion between the capping layer and the source/drain contact.
6. The device of claim 1, wherein the air spacer has width in a range of about 1 nanometer to about 5 nanometers.
7. The device of claim 1, wherein the air spacer has height in a range of about 5 nanometers to

about 40 nanometers.
8. The device of claim 1, comprising a spacer layer in contact with the gate structure, wherein the spacer layer has different etch selectivity than the capping layer.
9. The device of claim 1, comprising a gate via extending through the capping layer to an upper surface of the gate structure.
10. The device of claim 1, comprising: an etch stop layer between the gate structure and the source/drain contact, and between the air spacer and the source/drain region.
11. A device comprising: a substrate; a gate structure wrapping around at least one vertical stack of nanostructure channels over the substrate; a first source/drain contact over the substrate; a second source/drain contact laterally offset from the first source/drain contact; a capping layer laterally between the first and second source/drain contacts; a first air spacer positioned laterally between the first source/drain contact and the capping layer; and a second air spacer positioned laterally between the second source/drain contact and the capping layer, wherein each of the first and second air spacers is entirely located within a vertical region defined by a lower boundary that is at or above a top surface of the gate structure and an upper boundary that is at or below a top surface of the capping layer.
12. The device of claim 11, comprising: a first etch stop layer between the first source/drain contact and the first air spacer; and a second etch stop layer between the second source/drain contact and the second air spacer.
13. The device of claim 12, comprising: a third etch stop layer overlying the first and second air spacers, including: a first seal portion laterally between the capping layer and the first etch stop layer; and a second seal portion laterally between the capping layer and the second etch stop layer.
14. The device of claim 13, wherein the first seal portion has height less than about 10 nanometers.
15. The device of claim 11, comprising: a first spacer layer in contact with the gate structure and the capping layer; and a second spacer layer in contact with the gate structure and the capping layer; wherein upper surfaces of the gate structure, the first spacer layer and the second spacer layer are substantially coplanar.
16. A method, comprising: forming a gate structure over and wrapping around a vertical stack of nanostructures over a substrate; forming a first capping layer on the gate structure; removing the first capping layer; forming a liner layer on vertical sidewalls of source/drain contacts on either side of the gate structure; forming a second capping layer on the gate structure between the liner layer; and forming air spacers by removing the liner layer.
17. The method of claim 16, comprising: sealing the air spacers by forming a first etch stop layer over the second capping layer, the source/drain contacts and the air spacers.
18. The method of claim 17, comprising: recessing spacer layers on either side of the gate structure prior to forming the liner layer.
19. The method of claim 18, comprising: recessing second etch stop layers on either side of the gate structure prior to forming the liner layer.
20. The method of claim 16, wherein the forming a second capping layer includes: forming a second capping layer having lower dielectric constant than that of the first capping layer.